



CHALLENGE BASED MODULE

DECARBONIZATION OF TRANSPORT BY USE OF E-FUELS

Regulatory and Market Analysis of eSAF

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LEVEL 1: EXECUTIVE SUMMARY

The decarbonisation of aviation presents one of the most challenging aspects of the energy transition due to the continued dependence of aircraft on high energy-density liquid fuels. Electro-Sustainable Aviation Fuels (eSAF) have emerged as a promising long-term solution, particularly through Power-to-Liquid pathways such as Fischer-Tropsch (FT) and Methanol-to-Jet (MTJ). However, the deployment of these fuels depends not only on technological feasibility, but also on regulatory support, market development, economic competitiveness, and industrial readiness.

This work first reviews the technical and certification framework governing synthetic aviation fuels. ASTM D7566 standards and the current status of FT-SPK and MTJ pathways are examined, together with the broader international context provided by the CORSIA framework. These certification requirements determine the commercial readiness of alternative eSAF pathways and influence their potential contribution to aviation decarbonisation.

The regulatory framework is then analysed through the ReFuelEU Aviation Regulation. By introducing legally binding SAF blending mandates and dedicated eSAF sub-targets, the regulation creates a long-term demand signal intended to stimulate investment throughout the value chain. In addition to defining eligible fuels and stakeholder obligations, the regulation seeks to establish the conditions necessary for the emergence of a European eSAF market.

Building upon this regulatory framework, the mechanisms enabling market development and the limitations associated with implementation are investigated. Demand projections indicate a substantial increase in eSAF requirements over the coming decades. However, current production capacities remain insufficient to satisfy future targets, creating a significant mismatch between regulatory ambitions and industrial capability.

The market and economic analysis highlight the present state of supply, fuel prices, and cost competitiveness. Attention is given to the major contributors to eSAF production costs, including renewable electricity, green hydrogen production, carbon sourcing, and capital expenditure. These factors strongly influence the economic viability of large-scale deployment.

Finally, the analysis examines the transition from policy ambitions to industrial reality. Key barriers are identified, including infrastructure requirements, capital intensity, renewable energy availability, carbon dioxide sourcing, and broader economic and systemic risks. These constraints demonstrate that achieving long-term ReFuelEU targets requires more than regulatory support alone; it requires the simultaneous development of an integrated industrial ecosystem capable of producing synthetic fuels at commercial scale.

Overall, the analysis reveals that while strong regulatory drivers and increasing market demand support the development of eSAF, significant economic and structural challenges remain. The successful deployment of synthetic aviation fuels will therefore depend on the

coordinated evolution of technology, infrastructure, financing, and policy, providing the context within which the Fischer-Tropsch and Methanol-to-Jet pathways are assessed.

LEVEL 2: DEEP DIVE

CHAPTER 1

WHAT IS AVIATION FUEL?

Aviation turbine fuel, commonly known as Jet-A1, is a highly refined kerosene-based fuel used to power commercial aircraft. Beyond energy density, aviation fuel must meet all the stringent requirements for safety, reliability, and compatibility with existing aircraft engines and fuel-handling infrastructure. These requirements cannot be bypassed by any alternative fuel regardless of its sustainability credentials.

The primary standard governing these requirements is ASTM D7566, the *Standard Specification for Aviation Turbine Fuel Containing Synthesized Hydrocarbons*, developed by ASTM International. ASTM International is a globally recognized standards organization that develops voluntary consensus technical standards across many industries, including aviation fuel (ASTM, 2023).

1.1 ASTM D7566 Certification and eSAF

ASTM D7566, *Standard Specification for Aviation Turbine Fuel Containing Synthesized Hydrocarbons*, is the primary standard governing the production, composition, and certification of synthetic aviation fuel blending components (ASTM, 2023). It specifies the chemical and physical properties, performance criteria, and approved blending limits required to ensure that synthesized fuels are compatible with conventional aviation turbine fuels and existing aircraft systems. Synthetic fuels approved under ASTM D7566 are certified as “drop-in” blending components, meaning they can be integrated into current aviation fuel infrastructure and used in existing aircraft engines without requiring modifications.

Conventional aviation turbine fuels used in aircraft operations are governed by ASTM D1655, which is the main specification for Jet A and Jet A-1 fuels. Synthetic fuels certified under ASTM D7566 must first be blended with conventional ASTM D1655-compliant fuel within the approved blend limits before the final blended fuel itself becomes compliant with ASTM D1655 for operational use. Current approved synthetic pathways generally allow blending up to 50% synthetic fuel with conventional Jet A/Jet A-1 fuel (ASG Analytik, 2023). This blending requirement exists primarily because many synthetic fuels contain very low aromatic content, whereas a minimum aromatic fraction remains important for maintaining seal compatibility and material integrity in existing aircraft fuel systems.

ASTM therefore plays a critical role in defining the technical, compositional, and safety requirements necessary to ensure that synthetic aviation fuels can be used safely, reliably, and compatibly within existing aircraft engines and fuel-handling infrastructure (NREL, 2024).

A summary of key aviation fuel properties and the certification status of the main eSAF pathways is provided in Table 1 below.

Table 1: Key Aviation Fuel Properties and eSAF Pathway Certification Status

Property	Jet-A1 (Conventional)	FT-SPK (Fischer-Tropsch)	MtJ (Methanol-to-Jet)
Energy density (MJ/kg)	~43.2	~43.9	~42.8
Freezing point	$\leq -47^{\circ}\text{C}$	$\leq -47^{\circ}\text{C}$	$\leq -47^{\circ}\text{C}$
Flash point	$\geq 38^{\circ}\text{C}$	$\geq 38^{\circ}\text{C}$	$\geq 38^{\circ}\text{C}$
Aromatic content (% vol)	8–25%	<0.5% (neat)	Variable
Sulfur content	$\leq 0.30\%$	Near zero	Near zero
Max blend limit	—	50%	50% (proposed)
ASTM D7566 approval	Reference standard	Approved (Annex A1)	Pending
ReFuelEU eligible	Yes	Yes	Not yet

FT-SPK is currently an ASTM D7566-approved synthetic aviation fuel pathway, whereas Methanol-to-Jet (MtJ) remains under the ASTM certification process. ASTM approval is important because it determines whether a fuel pathway can be safely used within existing aircraft and fuel infrastructure under approved blending conditions. However, ASTM D7566 certification and ReFuelEU eligibility are not strictly equivalent. In addition to technical fuel certification, ReFuelEU eligibility also depends on compliance with sustainability and lifecycle greenhouse gas reduction criteria defined under the Renewable Energy Directive (RED II / RED III). Consequently, a fuel pathway may achieve ASTM approval while still requiring separate assessment against EU sustainability and emissions criteria before qualifying toward ReFuelEU aviation targets.

1.2 Approved eSAF Technology: Fischer-Tropsch (FT-SPK)

Among eSAF pathways, Fischer-Tropsch synthesized paraffinic kerosene (FT-SPK) is currently the only ASTM-approved electro-synthetic pathway under Annex A1 of D7566. In this process, green hydrogen (produced via electrolysis) is combined with captured CO₂ to generate syngas (CO + H₂), which is then catalytically converted to long-chain hydrocarbons and upgraded to jet-range paraffins. FT-SPK exhibits properties closely matching conventional kerosene, including low sulfur content, thermal stability, and suitable paraffinic profiles, allowing it to be blended safely with conventional jet fuel up to 50% (ASTM, 2023; NREL, 2024).

1.3 Methanol-to-Jet (MTJ) Pathway and Ongoing Certification

The Methanol-to-Jet (MTJ) pathway represents an emerging eSAF route: CO₂ and hydrogen are first converted to methanol, which is subsequently transformed through olefin into jet-range hydrocarbons. While promising due to its feedstock flexibility and potential scalability, MTJ is not yet approved under ASTM D7566 (Dolan, 2023). While promising due to its feedstock flexibility and potential scalability, MTJ is not yet approved under ASTM D7566, as it is still undergoing the rigorous ASTM D4054 qualification process which requires comprehensive demonstration of fuel composition consistency, performance data, and engine compatibility testing (ASTM, 2023; NREL, 2024). ASTM approval is anticipated in the 2026–2027 timeframe (Argus Media, 2026). Several companies, including ExxonMobil, Honeywell UOP, and Haldor Topsoe, are actively developing MTJ technologies and pushing the pathway through the ASTM approval process. Licensors like Honeywell UOP develop the technology and license it to producers for commercial deployment, earning fees from users (Dolan, 2023).

1.4 CORSIA framework

The Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA) is a global market-based mechanism established by the International Civil Aviation Organization (ICAO) to address carbon dioxide emissions from international aviation. CORSIA was adopted by ICAO Member States during the 39th ICAO Assembly in October 2016 and represents the first global sector-wide climate mechanism specifically designed for international aviation (ICAO, 2016). The scheme was developed in recognition of the rapidly increasing contribution of aviation to global greenhouse gas emissions and the need for a coordinated international approach to aviation decarbonization. Aviation faces significant challenges in reducing emissions because aircraft require fuels with high energy density, while large-scale electrification and hydrogen-powered aviation remain technologically limited for long-haul operations. Consequently, ICAO identified a combination of technological improvements, operational efficiencies, sustainable aviation fuels, and market-based measures as necessary components of the sector's long-term decarbonization strategy.

CORSIA was designed to complement these efforts by addressing the growth in international aviation emissions. Rather than imposing direct emissions caps or fuel mandates, the scheme requires airlines to compensate for a portion of their emissions through carbon offsetting and the use of eligible lower-carbon aviation fuels. The underlying principle is that while aviation

emissions may continue to grow in the short term, airlines should take responsibility for mitigating the climate impact associated with that growth (ICAO, 2024a).

The scheme applies exclusively to international flights operating between participating states. Domestic aviation emissions are excluded from CORSIA and remain under the jurisdiction of national governments. ICAO adopted this approach because international aviation operates across multiple jurisdictions, making coordinated global regulation more appropriate than fragmented national measures. The scheme therefore provides a common framework for participating countries while maintaining consistency across the international aviation market.

A fundamental component of CORSIA is its Monitoring, Reporting and Verification (MRV) system. Aircraft operators are required to monitor fuel consumption and calculate associated CO₂ emissions from international flights using approved methodologies. These emissions data are then independently verified and submitted to national authorities and ICAO. The MRV framework is intended to ensure transparency, consistency, and credibility in emissions accounting across participating states (ICAO, 2024b). Reliable emissions monitoring is essential because airlines' future offsetting obligations depend directly on the accuracy of reported emissions data.

A central feature of CORSIA is its offsetting mechanism. Airlines operating covered international routes may be required to purchase eligible emissions units from approved carbon reduction projects to compensate for a portion of their emissions. These projects may include activities such as renewable energy deployment, forest conservation, methane capture, or other initiatives that generate verified emissions reductions. The purpose of the offsetting mechanism is to ensure that emissions growth in aviation is balanced by emissions reductions achieved elsewhere in the global economy. ICAO has established criteria for the approval of emissions units to maintain environmental integrity and avoid double counting of emissions reductions (ICAO, 2024c).

Sustainable Aviation Fuels play an increasingly important role within CORSIA. The scheme recognizes that aviation can reduce its emissions not only through offsetting but also by directly lowering the carbon intensity of the fuel used. Airlines that utilize CORSIA-eligible fuels may claim lifecycle greenhouse gas emissions reductions associated with those fuels, thereby reducing their offsetting requirements. This creates an economic incentive for airlines to adopt SAF and, in the future, electro-Sustainable Aviation Fuels (eSAF). To qualify under CORSIA, fuels must satisfy specific sustainability criteria and demonstrate lifecycle emissions reductions relative to conventional fossil jet fuel (ICAO, 2024d).

CORSIA remains the principal global framework for addressing emissions from international aviation. By combining emissions monitoring, carbon offsetting, sustainability criteria for aviation fuels, and international cooperation, the scheme provides an important mechanism for reducing aviation's climate impact while supporting the transition toward lower-carbon fuels.

1.5 Carbon Border Adjustment Mechanism (CBAM)

The Carbon Border Adjustment Mechanism (CBAM) is an EU climate policy instrument designed to address the risk of carbon leakage by placing a carbon cost on certain carbon-intensive goods imported into the European Union. Carbon leakage occurs when production is relocated from the EU to countries with less stringent climate policies, or when EU production is displaced by more carbon-intensive imports. CBAM was established by Regulation (EU) 2023/956 and is designed to complement the EU Emissions Trading System (EU ETS) by extending a comparable carbon-cost signal to covered imports (European Parliament and Council of the European Union, 2023).

The mechanism forms part of the EU's broader decarbonisation strategy. Under the EU ETS, European producers in covered sectors are progressively exposed to the cost of their greenhouse-gas emissions. Historically, free allocation of EU ETS allowances has been used to reduce the risk of carbon leakage. CBAM is intended to replace this protection progressively, so that imported products face a carbon cost corresponding to their embedded emissions as the free allocation of EU ETS allowances is phased out. In this way, CBAM seeks to maintain more comparable carbon-cost conditions between EU production and imports while encouraging emissions reductions both inside and outside the EU (European Commission, 2026a).

Development and implementation of CBAM

CBAM was introduced through a transitional period from October 2023 to December 2025. During this phase, importers were required primarily to report the greenhouse-gas emissions embedded in covered imports; they did not yet have to purchase and surrender CBAM certificates. The transitional period was intended to establish the necessary systems for monitoring, calculating and reporting embedded emissions before the financial component of the mechanism became operational (European Commission, 2026a).

The definitive CBAM regime entered into force on 1 January 2026. Under the current system, EU importers of covered goods must become authorised CBAM declarants when they exceed the applicable threshold, report the embedded emissions associated with their imports and surrender CBAM certificates corresponding to those emissions. Following amendments adopted in 2025, a single mass-based threshold of 50 tonnes of CBAM goods per year applies to importers, excluding electricity and hydrogen from this mass threshold. The simplification is intended to remove very small importers from the administrative burden while retaining coverage of the vast majority of emissions within the mechanism (European Commission, 2025; European Commission, 2026a).

The price of CBAM certificates is linked directly to the EU carbon market. In 2026, certificate prices are calculated as quarterly averages of EU ETS auction prices, while from 2027 onwards the price will be calculated on a weekly basis. Where a carbon price has already been effectively paid in the country of production, the corresponding amount can be deducted from the CBAM obligation, thereby reducing the risk of the same emissions being subject to two carbon charges (European Commission, 2026a; European Commission, 2026b).

For imports made during 2026, the first CBAM declaration and surrender of certificates will take place in 2027. Importers will have to declare the embedded emissions associated with their 2026 imports and surrender the corresponding certificates by the applicable deadline. The mechanism therefore creates a direct financial incentive for producers outside the EU to reduce the carbon intensity of goods supplied to the European market (European Commission, 2026c).

Goods currently covered by CBAM

CBAM currently applies to selected goods in six sectors:

- Cement
- Iron and steel
- Aluminium
- Fertilisers
- Electricity
- Hydrogen

These sectors were selected because of their relatively high emissions intensity and exposure to carbon-leakage risks (European Commission, 2026a).

The inclusion of hydrogen is particularly relevant to the eSAF value chain. eSAF produced through a Power-to-Liquid (PtL) pathway requires substantial quantities of hydrogen, normally produced through electrolysis when renewable hydrogen is used. Consequently, the carbon intensity and origin of hydrogen can influence the overall environmental and economic performance of eSAF production. CBAM creates a carbon-cost mechanism for certain imported hydrogen products, potentially affecting the relative economics of importing hydrogen into the EU compared with producing lower-carbon hydrogen within the EU.

Relevance of CBAM to eSAF

It is important to distinguish between direct and indirect relevance. CBAM does not currently impose a carbon border charge directly on aviation fuel or eSAF. Aviation fuel is not one of the six sectors presently covered by CBAM. Therefore, it would be inaccurate to describe CBAM as a direct regulatory mechanism for eSAF.

Its relevance arises primarily through the inputs and industrial systems surrounding eSAF production. A typical eSAF PtL pathway requires renewable electricity, hydrogen and a sustainable source of carbon dioxide. Hydrogen is particularly important because its production can represent a major component of the energy requirement and production cost of eSAF. Where hydrogen is imported in a form covered by CBAM, its embedded emissions may create an additional carbon-related cost at the EU border. This can improve the relative competitiveness of lower-carbon hydrogen compared with hydrogen produced using carbon-intensive processes (European Commission, 2026a).

CBAM can therefore influence the upstream economics of eSAF, rather than directly regulating the final aviation fuel. For example, an eSAF producer using low-carbon hydrogen would potentially face a different cost structure from a producer relying on hydrogen with a higher embedded carbon intensity. The mechanism consequently reinforces the economic value of reducing emissions in energy-intensive upstream processes.

CBAM is also relevant to the electricity-intensive nature of eSAF production. PtL production requires large amounts of renewable electricity for hydrogen electrolysis and other conversion processes. Electricity imported into the EU is already within the current CBAM scope. Although this does not mean that electricity consumed by an EU-based eSAF plant is itself subject to CBAM, the mechanism contributes to the wider EU policy environment in which carbon-intensive imported electricity is assigned a carbon cost. This reinforces the broader EU objective of reducing the carbon intensity of energy-intensive industrial production (European Commission, 2023).

CBAM and the competitiveness of European eSAF

One of the major challenges facing eSAF is its substantially higher production cost compared with conventional fossil-based aviation fuel. The economics of eSAF therefore depend not only on the cost of renewable electricity, hydrogen and CO₂, but also on the regulatory treatment of competing energy and fuel production pathways.

CBAM can contribute to a more level competitive environment by incorporating the carbon intensity of certain imported products into their effective cost. Without such a mechanism, European producers facing carbon costs through the EU ETS could potentially compete against imports produced under less stringent carbon constraints. CBAM is designed to reduce this disparity by requiring importers of covered goods to account financially for their embedded emissions (European Parliament and Council of the European Union, 2023).

For eSAF, this is particularly relevant to the development of European hydrogen and Power-to-Liquid value chains. If European eSAF projects rely on low-carbon hydrogen while competing with imported carbon-intensive hydrogen or other energy-intensive inputs, the carbon-cost signal provided by CBAM can alter the relative economics of those supply options. However, CBAM alone does not make eSAF cost-competitive with fossil jet fuel. The mechanism is therefore better understood as one component of a wider policy framework that includes ReFuelEU Aviation, the EU ETS, the Innovation Fund, renewable-energy policy and financial support mechanisms.

Relationship between CBAM and the EU ETS

CBAM and the EU ETS are closely connected but perform different functions. The EU ETS primarily places a carbon price on emissions generated by covered activities within the EU, whereas CBAM addresses the embedded emissions of specified goods imported from outside the EU. The two mechanisms are designed to operate together rather than independently.

This relationship is particularly important because CBAM is being phased in alongside the gradual reduction of free EU ETS allowances for sectors covered by the mechanism. The

CBAM obligation therefore increases as the corresponding protection provided by free allocation decreases. Full integration with the phase-out of free allocation is planned over the period to 2034 (European Commission, 2023; European Commission, 2026a).

This structure is intended to maintain the environmental effectiveness of EU climate policy while reducing incentives to relocate carbon-intensive production outside the EU.

Calculation of embedded emissions

A central feature of CBAM is that the importer is not simply charged a fixed tariff on the physical quantity of a product. Instead, the mechanism is based on the greenhouse-gas emissions embedded in the imported goods. Importers must therefore determine the emissions associated with the production of the relevant goods.

Actual emissions data can be used where the necessary information is available and appropriately verified. The European Commission has established methodologies and default values to support the calculation of embedded emissions where appropriate. For actual emissions, non-EU installation operators provide emissions information that can be subject to verification by accredited CBAM verifiers (European Commission, 2026d).

This monitoring and verification requirement is significant for international supply chains. Producers outside the EU that wish to demonstrate a lower carbon intensity for their products increasingly need reliable emissions accounting and data systems. Consequently, CBAM does not only create a financial obligation; it also encourages greater carbon transparency and emissions monitoring throughout international industrial supply chains.

Future development and relevance to the eSAF sector

The scope of CBAM is not necessarily static. The European Commission has been examining potential extensions of the mechanism, including the treatment of indirect emissions and additional sectors. In 2026, the Commission published a technical study examining how indirect emissions could be incorporated more extensively, including questions concerning electricity-related emissions, actual emissions claims, Power Purchase Agreements (PPAs), verification and the potential extension of indirect-emissions coverage to additional CBAM sectors (European Commission, 2026e).

This development could become relevant to energy-intensive eSAF value chains because PtL production depends heavily on electricity and hydrogen. However, any future extension should not be assumed to apply automatically to eSAF. Such changes would require further EU legislation and methodological development.

Overall, CBAM should therefore be viewed as an indirect but strategically relevant policy instrument for eSAF. It does not directly regulate eSAF production or impose a border charge on aviation fuel. Instead, it contributes to the EU's wider carbon-pricing architecture by placing a carbon cost on selected carbon-intensive imports, including hydrogen, while supporting the transition away from free EU ETS allowances. For the emerging eSAF industry, its principal significance lies in shaping the competitiveness and carbon intensity of upstream inputs such as hydrogen and electricity and in encouraging more transparent carbon

accounting across international supply chains. Its effectiveness for eSAF will ultimately depend on how the mechanism evolves alongside ReFuelEU Aviation, the EU ETS and the broader EU strategy for renewable hydrogen and industrial decarbonisation.

CHAPTER 2

SUSTAINABLE AIR FUEL: THE REFUEL EU PERSPECTIVE

This chapter explores the technical, regulatory, and market dimensions of eSAF deployment under ReFuelEU, highlighting the criteria, approved fuel pathways, market trends, and the challenges that must be addressed to scale eSAF globally.

2.1 Regulatory Analysis: RefuelEU Regulation

Aviation accounts for a significant share of global greenhouse gas emissions, with limited near-term alternatives to conventional jet fuel. To achieve net-zero targets and reduce the sector's climate impact, sustainable aviation fuels (SAFs) have emerged as a critical solution. Among these, electro-synthetic aviation fuels (eSAF) produced from renewable hydrogen and captured CO₂ via power-to-liquid (PtL) processes, offer the potential for deep decarbonization while remaining compatible with existing aircraft and fuel infrastructure (IEA, 2023; NREL, 2024).

The European Union has introduced regulatory frameworks to accelerate SAF adoption, with ReFuelEU Aviation being one of the most significant initiatives. Launched under the European Green Deal, ReFuelEU aims to mandate a minimum blending of SAF into conventional jet fuel at EU airports, incentivize the production and deployment of SAF, and stimulate investment in low-carbon fuel technologies (European Commission, 2023). The ReFuelEU Aviation Regulation plays a central role in accelerating the decarbonisation of the aviation sector within the European Union. Within this context, the ReFuelEU initiative provides a market signal and policy driver for emerging eSAF technologies, including Fischer-Tropsch (FT-SPK) and Methanol-to-Jet (MTJ) pathways. By combining regulatory mandates, technical standards, and economic incentives, the framework seeks to bridge the gap between demonstration-scale production and industrial-scale deployment of eSAF, positioning Europe as a global leader in sustainable aviation (IEA, 2023; Dolan, 2023).

The regulation works primarily through **fuel supply mandates**. Aviation fuel suppliers operating at covered EU airports must ensure that the fuel they place on the market contains minimum shares of SAF, increasing progressively over time. This creates legally guaranteed demand for SAF and provides a stronger investment signal to producers than voluntary market uptake alone (European Commission, 2023). The mandated minimum SAF shares are given in the table below.

Table 2: Mandated Minimum SAF Shares according to the RefuelEU regulation

Year	Minimum SAF Share
2025	2%
2030	6%
2035	20%
2040	34%

2045	42%
2050	70%

Within these overall SAF targets, the regulation also creates a specific sub-mandate for synthetic aviation fuels, effectively eSAF. This is significant because without a dedicated quota, lower-cost bio-based pathways might dominate compliance while synthetic fuels remain commercially marginal.

Table 4: Mandated Minimum eSAF Shares according to the RefuelEU regulation

Year	Minimum eSAF Share
2030	1.2%
2035	5%
2040	10%
2045	15%
2050	35%

2.2 SAF Eligibility Criteria

The regulation sets out specific criteria that a fuel must satisfy across multiple dimensions, feedstock origin, energy and lifecycle greenhouse gas performance, before it can count toward blending targets.

1. Feedstock criteria: SAF must be produced from feedstocks that are explicitly recognised under the Renewable Energy Directive (RED III). These include:

- Agricultural and forestry residues (e.g. straw, bagasse, sawdust)
- Municipal solid waste and non-recyclable waste fractions
- Used cooking oil and certain animal fats (not competing with food supply)
- Renewable non-biological sources, specifically renewable electricity and water, used to produce green hydrogen for eSAF via PtL

Importantly, feedstocks that compete with food and feed crops (first-generation biofuels) are explicitly excluded. This is a deliberate policy choice to avoid the land-use conflicts that undermined the credibility of early biofuel mandates.

2. Electricity criteria: For eSAF specifically, the electricity used in the electrolysis process must qualify as a Renewable Fuel of Non-Biological Origin (RFNBO) under RED III. This means the green hydrogen feedstock must be produced from renewable electricity that is additional, temporally correlated, and geographically correlated with the renewable

generation source. This strict additional criterion is designed to ensure the hydrogen is genuinely low carbon rather than simply drawing from the general grid.

3. GHG savings criteria: SAF must deliver a minimum lifecycle greenhouse gas saving of 65% compared to the fossil fuel. Lifecycle emissions are assessed using the methodology set out in RED III, covering feedstock production, processing, transport, and combustion.

Compliant solutions under the legislation criteria are summarised in the table below.

Table 5: SAF Technology Categories and ReFuelEU Eligibility.

SAF Category	Feedstock	GHG Saving (vs fossil)	ReFuelEU Eligible
Advanced biofuels	Agricultural/forestry residues	65–85%	Yes
HEFA (used cooking oil)	Used cooking oil, animal fats	60–80%	Yes (with limits)
Recycled Carbon Fuels	Non-recyclable waste, flue gas	55–70%	Yes
eSAF / PtL (FT-SPK)	Green H ₂ + captured CO ₂	85–95%+	Yes (ASTM approved)
eSAF / PtL (MtJ)	Green H ₂ + CO ₂	85–95%+	Not yet (pending ASTM)
First-gen biofuels	Food/feed crops	Variable	Excluded

2.3 Key stakeholders:

The successful implementation of ReFuelEU aviation requires the collaboration of all parties in the supply chain, in particular aviation fuel suppliers, EU airports and airlines. More than 95% of air transport departing from EU airports will be covered by this new regulation.

- Aviation fuel suppliers at EU airports** will gradually increase the share of SAF blended with conventional aviation fuel, ensuring the fuel sold at EU airports contains the required SAF share. Example: If a supplier sells 1 million tonnes of jet fuel in 2030, 6% must be SAF.
- Aircraft operators departing from EU airports** must refuel with the aviation fuel necessary to operate the flight. This avoids the excessive emissions related to extra weight and minimises the risks of carbon leakage caused by ‘tankering’ practices (article 5). Airlines must refuel at least 90% of their required fuel at EU airports. Fuel tankering = airlines loading extra fuel outside the EU to avoid SAF mandates.

3. **EU airports** must facilitate access to the necessary infrastructure to deliver, store and refuel aircraft with SAF (article 6). Airports must provide SAF infrastructure and logistics so suppliers can deliver the mandated fuels.

Aircraft operators departing from EU airports must refuel with the aviation fuel necessary to operate the flight, with airlines required to refuel at least 90% of their required fuel at EU airports to minimise carbon leakage caused by tankering practices (Article 5). EU airports must facilitate access to the necessary infrastructure to deliver, store, and refuel aircraft with SAF (Article 6). Aviation fuel suppliers, EU airports, and aircraft operators are required to collect data and report to the European Union Aviation Safety Agency (EASA) and competent authorities to monitor compliance (European Commission, 2023).

2.4 Obligations of Key Stakeholders:

Aviation fuel suppliers must ensure that the aviation fuel blend made available to aircraft operators at EU airports contains a minimum share of SAF from 2025 and a minimum share of synthetic aviation fuels from 2030. Both shares will increase progressively until 2050. SAF must be compliant with the renewable energy directive sustainability and emissions-saving criteria to count to the targets, and SAF includes:

- aviation biofuels, notably advanced biofuels and other biofuels, produced from waste and residues.
- synthetic aviation fuels, produced from renewable hydrogen (eSAF)
- recycled carbon aviation fuels.

Additionally, aviation fuel suppliers may choose to meet both minimum shares with hydrogen for direct use in aircraft (renewable and non-fossil low-carbon hydrogen), along with synthetic low-carbon aviation fuels (produced from non-fossil low-carbon hydrogen).

Aircraft operators (airlines) departing from EU airports must refuel with the aviation fuel necessary for their entire flight. This avoids the excessive emissions related to extra weight and minimising the risks of carbon leakage caused by tankering practices. Airlines must refuel at least 90% of their required fuel at EU airports. It should be noted, however, that the anti-tankering rule applies only to departures from EU airports, leaving open the possibility that airlines could strategically refuel at airports just outside EU jurisdiction, such as Istanbul or Casablanca, to reduce SAF exposure on return legs. This geographic limitation is a recognised structural gap in the regulation that the Commission has yet to fully address. EU airports managing bodies must facilitate access to the necessary infrastructure to deliver, store and refuel aircraft with SAF.

Aviation fuel suppliers, EU airports and aircraft operators are required to collect data and report to the European Union Aviation Safety Agency (EASA) and to competent authorities to monitor compliance with their obligations.

2.5 Geographical Scope and Non-EU Applicability

Although ReFuelEU Aviation is an EU regulation, aviation markets are international and highly interconnected. Beyond the EU, several neighbouring countries have adopted or are aligning with the regulation. Therefore, these non-EU countries and airports may be indirectly affected by the regulation, particularly where they compete with EU hubs for passenger traffic, cargo flows, and refuelling activity. Examples include the United Kingdom, Norway, Switzerland.

Airlines may compare fuel costs, SAF availability, airport charges, and operational flexibility across jurisdictions. If SAF compliance increases fuel costs at EU airports while neighbouring airports face lower obligations, there may be incentives for route restructuring, transfer traffic diversion, or fuel purchasing outside the EU system (Transport & Environment, 2024). Anti-tankering provisions were included in ReFuelEU, requiring airlines to uplift most required fuel at EU airports rather than systematically carrying extra fuel from cheaper non-EU locations (Regulation (EU) 2023/2405, Article 5).

Switzerland formally adopted ReFuelEU Aviation as of 1 January 2026, meaning fuel suppliers at Zurich and Geneva airports are now subject to the same blending obligations, ramping from 2% to 70% by 2050 (European Commission, 2026). Norway, while outside the EU, participates deeply in European aviation markets through the EEA framework and has historically adopted ambitious SAF policies, including blending mandates. The United Kingdom, following Brexit, has developed its own SAF mandate independently. Its SAF mandate requires 2% SAF in 2025 rising to 10% by 2030 and 22% by 2040, broadly comparable in ambition but with different sub-mandate structures and no specific eSAF sub-target equivalent to the EU's. (UK Department for Transport, 2024).

2.6 Penalty

The Regulation sets financial penalties for fuel suppliers and aircraft operators in case they fail to comply with the obligations. Different from a buy-out price in which you can buy yourself out of the obligation as a fuel supplier, suppliers and aircraft operators under ReFuelEU are faced with a penalty but still need to make up for their non-compliance in the following year. Income from penalties is required to be earmarked for funds that support SAF projects. Failing to comply with the minimum shares of SAF means fuel suppliers are liable to pay a fine of at least twice the difference between the price of conventional aviation fuel and SAF in that year. The European Aviation Safety Agency (EASA) is tasked with collecting market intelligence on SAF pricing every year. For example, if Jet-A1 price is €600/mt and SAF at €3,400/mt. The difference multiplied by two is €5,600 and this represents the amount the supplier will pay as a penalty.

2.7 Application Timeline:

The ReFuelEU Aviation Regulation was formally adopted in October 2023 and entered into force shortly after its publication in the Official Journal of the European Union. The regulation applies from 1 January 2024, marking the beginning of its legal effect across EU Member States. However, the core operational provisions, particularly those related to fuel

blending obligations apply from 1 January 2025, allowing stakeholders time to prepare for compliance (Council of the EU, 2023).

2024 was to serve primarily as a monitoring and reporting phase, during which aviation fuel suppliers and aircraft operators were required to begin data collection and reporting activities in preparation for full implementation. From 2025 onwards, the regulation becomes fully operational, with mandatory SAF blending targets taking effect, starting at 2% SAF in aviation fuel supplied at EU airports and increasing progressively toward long-term targets up to 2050 (European Commission, 2023). This phased implementation approach ensures a gradual transition, giving industry actors sufficient time to adapt infrastructure, supply chains, and operational practices before being subject to binding obligations.

2.8 Benefit of the Regulation

1. Contribution to EU Climate Targets: One of the primary benefits of ReFuelEU Aviation is its direct contribution to the European Union's climate objectives under the European Green Deal and the "Fit for 55" package. Aviation is a hard-to-abate sector with limited alternatives to liquid fuels, and eSAF represents one of the most viable solutions for reducing lifecycle greenhouse gas emissions. Sustainable aviation fuels can achieve lifecycle emission reductions ranging from approximately 65% to near 100%, depending on the production pathway and feedstock used (European Commission, 2023). By mandating increasing shares of eSAF, the regulation ensures a gradual but steady reduction in aviation-related emissions across the EU. (European Commission 2023)

2. Encouraging Investment and Industrial Development: The introduction of mandatory SAF blending targets is expected to stimulate large-scale investments in production facilities, supply chains, and supporting infrastructure. By creating long-term, legally binding demand, the regulation reduces revenue uncertainty for producers and provides a basis for project financing. However, this investment stimulus effect is not automatic. As explored further in Chapter 3, the business case for eSAF remains economically challenging, production costs are currently 8 to 13 times higher than conventional jet fuel, and no large-scale PtL project in the EU has yet reached a final investment decision. According to the European Commission, achieving SAF targets will require substantial capital investment in new production plants, particularly for advanced biofuels and synthetic fuels (European Commission, 2023). This creates opportunities for: development of new industrial value chains, expansion of hydrogen and CO₂ capture industries, integration with renewable energy systems.

3. Promotion of Technological Innovation: The regulation encourages innovation across multiple technological domains. While current SAF supply is dominated by biofuels such as HEFA, future targets, particularly the sub-targets for synthetic aviation fuels, drive the development of more advanced technologies such as: Power-to-Liquid (PtL) pathways, hydrogen production via electrolysis and carbon capture and utilisation. By creating guaranteed demand, ReFuelEU incentivises research, development, and scale-up of these technologies, which are essential for achieving long-term decarbonisation (EASA, 2025).

4. Creation of a Level Playing Field: ReFuelEU Aviation ensures that all airlines operating within the EU are subject to the same SAF requirements, thereby creating a level playing

field. Without such regulation, airlines using SAF would face higher operational costs compared to competitors relying solely on fossil fuels. The regulation eliminates this competitive disadvantage by making SAF use mandatory across the board. Additionally, measures such as the anti-tankering rule prevent airlines from bypassing SAF requirements by refuelling outside the EU, ensuring fair competition and preventing carbon leakage (European Commission, 2023).

5. Job Creation and Economic Growth: The expansion of SAF production and infrastructure is expected to generate new employment opportunities across multiple sectors, including: renewable energy, chemical engineering and fuel production and logistics. The development of SAF value chains also contributes to regional economic growth and industrial diversification within the EU (European Commission, 2023).

6. Supporting the Development of Global SAF Markets: By taking a leading role in SAF regulation, the EU sets a precedent for other regions. ReFuelEU Aviation contributes to the development of global SAF markets by establishing regulatory standards, encouraging international investment and promoting cross-border supply chains. This leadership role is particularly important for scaling eSAF, which requires global coordination in areas such as renewable energy production and CO₂ sourcing.

CHAPTER 3

3.1 Enablers for emergence of e-SAF market

ReFuelEU has successfully triggered a wave of project announcements. As of May 2025, 41 large-scale e-SAF projects are under development in Europe (up from 27 in January 2024), making it the region with the most projects in the world. These 41 projects have a potential combined production capacity of 2.8 million tonnes per year, close to three times the annual volume required by ReFuelEU by 2032 (Transport & Environment, 2025).

The development of eSAF is progressing through a series of technological, commercial, and infrastructural milestones. Unlike conventional SAF pathways, which are already commercially established, eSAF remains in an early stage of deployment. However, recent developments across Europe demonstrate clear progress from pilot-scale technologies toward commercialisation and system integration.

1. Commercial Project Development: A key milestone in eSAF deployment is the transition from pilot and demonstration phases to commercial project development. This stage involves critical activities such as site selection, renewable electricity sourcing, carbon dioxide supply, permitting, financing and engineering design. INERATEC has reached this milestone with the inauguration of its ERA ONE plant in Frankfurt, described as the largest commercial-scale power-to-liquid facility of its kind in Europe, with a production capacity of up to 2,500 tonnes of carbon-neutral e-fuels annually (GreenAir News, 2024). The plant produces synthetic crude oil from biogenic CO₂ and green hydrogen, which is then refined into eSAF, e-diesel, and other chemical products. Notably, the feedstocks are sourced locally within the Frankfurt-Höchst industrial park, where CO₂ is captured from a biogas plant and green hydrogen, which is then refined into eSAF, e-diesel, and other chemical products. The project is supported by a €70 million financing package from the European Investment Bank and Breakthrough Energy Catalyst (European Investment Bank, 2025). This milestone reflects growing investor confidence and indicates that eSAF technologies are moving beyond laboratory and pilot-scale validation.

2. Hydrogen Infrastructure and System Enablers

Beyond fuel production itself, the development of hydrogen infrastructure represents a critical enabling milestone for eSAF scale-up. Since hydrogen is a key input in Power-to-Liquid processes, its availability at scale is essential. Several approaches to hydrogen supply are under development across European industrial hubs, ranging from local electrolysis to import via chemical carriers. One such carrier route involves ammonia: hydrogen is converted to ammonia for transport and then cracked back to hydrogen at the point of use. Air Liquide has commissioned an industrial-scale ammonia cracking unit in Antwerp (Ammonia Energy Association, 2024), and the AmplifHy project in Rotterdam aims to establish hydrogen import terminals and cracking facilities for the same purpose (VTTI, 2024). It should be noted, however, that ammonia cracking as a hydrogen supply route carries significant efficiency penalties; the conversion of hydrogen to ammonia and back involves substantial energy losses, and the economics of this pathway are considerably weaker than

direct electrolysis where renewable electricity is available locally. These projects are therefore better understood as transitional infrastructure responses to a near-term hydrogen availability gap rather than long-term solutions.

3. Industrial Scale and Market Integration (Ongoing Challenge)

The final milestone in eSAF development is full industrial-scale deployment and integration into the aviation fuel market. This stage would involve multiple large-scale production plants, stable renewable hydrogen supply chains, scalable CO₂ sourcing, certified fuel pathways, and integration into airport fuel distribution systems. However, this milestone has not yet been fully achieved. Current market assessments indicate that synthetic fuels contribute only a negligible share of total SAF consumption, with the market still dominated by bio-based fuels (EASA, 2025). This suggests that while progress has been made in project development and early commercialisation, the eSAF sector remains positioned between initial market entry and largescale industrial deployment.

3.2 Pain points of the regulation

Despite its ambition and strategic importance, ReFuelEU Aviation has significant weaknesses. A few pain points have been outlined below and more will be discussed further in Chapter 4

1. The supply-demand timing mismatch is the most fundamental problem. The regulation creates legally binding demand from 2025 onwards, but the supply infrastructure needed to meet that demand such as PtL plants, electrolyzers, renewable energy capacity, CO₂ capture systems are complex to build, develop and deploy at scale. The regulation did not introduce parallel supply-side obligations or guaranteed financial help at the same time as the demand mandate.
2. The penalty structure creates perverse incentives. Penalties for non-compliance are set at twice the price differential between fossil jet fuel and SAF. While intended to be punitive, this creates a price ceiling rather than a compliance floor. A fuel supplier facing a penalty of €5,600 per tonne of shortfall may rationally choose to pay the fine rather than source eSAF at €8,700 per tonne particularly when supply is limited. The regulation risks becoming a fine collection mechanism rather than a genuine decarbonisation driver.
3. The geographic loophole undermines the level playing field. The anti-tankering rule applies only to departures from EU airports, meaning airlines can legally minimise SAF exposure by refuelling at border airports just outside EU jurisdiction, for example, in Istanbul.

CHAPTER 4

MARKET ANALYSIS OF ESAF IN THE EU

Within the broader SAF market, eSAF occupies a strategic long-term role, particularly in scenarios where biomass availability is limited. As such, the eSAF market is not yet fully established but is emerging as a policy-driven and innovation-dependent segment of the aviation fuel sector (European Commission, 2024).

4.1 eSAF Demand according to ReFuelEU Regulation Assumptions

To translate the percentage-based obligations of ReFuelEU Aviation into actual market fuel volumes, it is necessary to begin with an estimate of total aviation fuel demand in the European Union. The regulation sets blending mandates as percentages of total aviation fuel supplied at Union airports rather than fixed values. Consequently, the practical market size for SAF and eSAF can be derived by applying these percentages to the annual fuel demand.

For this market analysis, total EU aviation fuel demand is assumed at 46 million tonnes (Mt) per year. This assumption is consistent with European Commission modelling used in the ReFuelEU Aviation impact assessment and broader EU transport decarbonisation scenarios, where European aviation fuel demand remains in the range of approximately 40–50 Mt annually depending on traffic growth, efficiency improvements, and policy measures (European Commission, 2021; European Commission, 2024). The use of 46 Mt therefore represents a balanced mid-case scenario: it is sufficiently conservative to avoid overstating demand while remaining realistic for long-term planning purposes.

Under ReFuelEU Aviation, SAF shares must increase from 2% in 2025 to 70% by 2050. Within these overall SAF targets, the regulation also establishes a dedicated sub-target for synthetic aviation fuels (eSAF), increasing from 1.2% in 2030 to 35% by 2050.

Calculation Method

The required annual fuel volumes can be estimated by applying the mandated blending shares to the assumed annual aviation fuel demand.

Required Fuel Volume (Mt/year) = Total Aviation Fuel Demand (Mt/year) × Mandated Share

Accordingly:

SAF Demand (Mt/year) = 46 × SAF Target (%)

eSAF Demand (Mt/year) = 46 × eSAF Target (%)

Results

Applying the ReFuelEU Aviation blending mandates to the assumed annual fuel demand of 46 Mt/year yields the fuel volumes presented in Table X.

Table X: Estimated SAF and eSAF Demand Under ReFuelEU Aviation

Year	SAF Mandate	eSAF Mandate	Required SAF (Mt/year)	Required eSAF (Mt/year)
2025	2%	0%	0.92	0.00
2030	6%	1.2%	2.76	0.55
2035	20%	5%	9.20	2.30
2040	34%	10%	15.64	4.60
2045	42%	15%	19.32	6.90
2050	70%	35%	32.20	16.10

The results demonstrate the scale of fuel production required to satisfy future regulatory obligations. SAF demand increases from 0.92 Mt/year in 2025 to 32.2 Mt/year by 2050. More significantly, eSAF demand rises from 0.55 Mt/year in 2030 to 16.1 Mt/year in 2050.

These figures illustrate the magnitude of the industrial challenge facing the aviation sector. While the regulation creates a clear and predictable demand signal, achieving these volumes will require a substantial expansion of renewable electricity generation, hydrogen production capacity, CO₂ sourcing infrastructure, and eSAF production facilities across Europe.

4.2 Current Supply and Market Conditions

Despite strong policy-driven demand signals, the supply of eSAF remains extremely limited. According to EASA, SAF accounted for approximately 0.6% of total aviation fuel consumption in the EU in 2024, with the overwhelming majority derived from bio-based pathways such as HEFA (EASA, 2025). Synthetic fuels, including eSAF, contributed negligibly to total supply, highlighting a significant gap between regulatory targets and actual production capacity. Globally, SAF production reached approximately 1.3 million tonnes in 2023, representing less than 1% of total jet fuel demand, with eSAF contributing only a marginal share (International Energy Agency, 2024). In the EU, more than 40 e-fuel projects are currently in planning stages, yet very few have reached final investment decision (FID), indicating that the sector is still transitioning from pilot-scale demonstration to early commercialisation (European Commission, 2024).

For eSAF, high capital costs and high price and revenue risks currently undermine the business. More than 40 e-fuels production projects are at a planning stage in the EU. However, none has been able to reach a final investment decision so far. Different contracting needs present an obstacle for the market to take off. Fuel producers aim for long-term fuel offtake agreements to amortise costly investments, whilst operators are accustomed to short-term supply contracts to remain more flexible with respect to fuel pricing in competitive global markets. Reluctance to invest is further fuelled by uncertainties with respect to available biological and nonbiological feedstock and their overall access conditions.

4.3 Current Market Prices and Cost Competitiveness

The economic viability of eSAF is currently constrained by its significantly higher production costs compared to conventional jet fuel. According to the European Union Aviation Safety Agency (EASA) 2024 Aviation Fuels Reference Prices for ReFuelEU Aviation:

- Conventional jet fuel: ~€734/tonne
- SAF average (2024): ~€2,085/tonne
- PtL (eSAF) estimated cost: up to €8,700/tonne

This means eSAF can be 3–10 times more expensive than fossil jet fuel. As seen, the main hurdle for ramping up the market is the significant price difference between renewable and low-carbon fuels and conventional fuels. Using these fuels results in significantly higher operating costs for airlines and shipping operators. Costs are expected to come down in the coming years, due to economies of scale, technological learning and new production technologies. However, uncertainties surrounding expected price and revenue causes hesitation among investors.

4.4 Cost breakdown for eSAF Production

eSAF remains significantly more expensive than conventional jet fuel because it has a stack of interconnected upstream cost drivers, each of which carries its own capital and energy intensity challenges. It combines several capital-intensive and energy-intensive process steps: hydrogen production, carbon supply, fuel synthesis, upgrading, and downstream logistics. A detailed cost breakdown of each process is outlined below

Renewable Electricity: Renewable electricity is the foundational and largest cost driver across the entire eSAF production chain. It is consumed at every major stage powering the electrolysis process that produces green hydrogen, and in the case of Direct Air Capture, powering the CO₂ sourcing step as well. Electricity alone accounts for an estimated 45–55% of total eSAF production cost, making it the single most important lever for cost reduction over time (FTI Consulting, 2025; NREL, 2024).

Green Hydrogen Production: Green hydrogen produced via electrolysis of water using renewable electricity is the primary chemical feedstock of the PtL process and the second largest cost component. The levelized cost of green hydrogen in Europe depends on electrolyser technology, capacity factor, electricity price, and plant scale (European Hydrogen Observatory, 2024)

CO₂ Capture: The cost of CO₂ varies enormously depending on the sourcing route chosen, and this choice has a significant impact on total eSAF production cost. There are two main options in practice. The first is point-source capture where CO₂ is captured from an industrial facility that already produces concentrated CO₂ emissions, such as a biogas upgrading plant, cement plant, or waste incinerator. This is the cheaper and more immediately practical route (ICCT, 2022). The second route is Direct Air Capture (DAC), which extracts CO₂ directly from the atmosphere. While DAC is far more scalable, geographically flexible in the long run and produces eSAF with higher lifecycle GHG savings, it is significantly more expensive at

current technology maturity (ICCT, 2022; PMC, 2024). For near-term projects, point-source biogenic CO₂ is therefore the pragmatic default, with DAC representing a longer-term scaling pathway as costs decline.

Fuel Synthesis and Upgrading: Once green hydrogen and CO₂ are available, the actual fuel synthesis step, whether via Fischer-Tropsch reaction or Methanol-to-Jet conversion, represents a comparatively smaller share of total cost, estimated at 10–15% of total production cost (NREL, 2024). This is the most technically mature component of the PtL chain, benefiting from decades of industrial experience in syngas chemistry and refinery operations. Upgrading and fractionation of the synthetic crude into jet-range kerosene adds a further 5–8% of cost. While these steps are not negligible, they are not the primary economic barrier to eSAF deployment, the bottleneck lies firmly upstream in electricity and hydrogen costs.

Capital Expenditure: Beyond operating costs, the high capital expenditure (CAPEX) required to build PtL plants represents a significant additional barrier. eSAF production facilities are highly complex and capital intensive, integrating multiple emerging technologies into a single value chain (FTI Consulting, 2025). This capital intensity means that producers require long payback periods and revenue certainty to justify investment, which in turn drives the demand for long-term offtake agreements that airlines are currently reluctant to provide.

Future Cost Outlook

Costs are expected to decline progressively over the coming decades, driven by falling renewable electricity prices, electrolyser learning curves, and economies of scale in CO₂ capture. FTI Consulting projects eSAF production costs could fall to approximately \$5–\$9 per gallon by 2030, equivalent to roughly €1,500–€2,700 per tonne, representing a significant reduction from current levels but still well above fossil jet fuel parity (FTI Consulting, 2025). Even under optimistic projections, eSAF is unlikely to reach cost competitiveness with fossil kerosene without sustained carbon pricing, regulatory mandates, and direct financial support mechanisms.

CHAPTER 5

ECONOMICS OF ESAF IN THE EUROPEAN UNION

5.1 From Policy Targets to Industrial Reality

The ReFuelEU Aviation Regulation establishes legally binding targets for the incorporation of eSAF. While these targets provide a strong demand signal, they do not guarantee the availability of supply. The realisation of these targets depends on the ability to deploy large-scale industrial infrastructure within relatively short timeframes.

5.2 Market Constraints and Structural Barriers.

The eSAF market is characterised by numerous interconnected constraints that collectively limit its growth, spanning technical, economic, infrastructural, and logistical dimensions. The market constraints can be closely linked with the economic risks to be discussed in Chapter 4.

On the technical side, the limited availability of renewable hydrogen at the scale required for PtL production remains a fundamental challenge. Alongside this, challenges in sourcing sustainable CO₂ at sufficient volumes and the low maturity of large-scale PtL conversion technologies add further layers of technical constraints.

The economic constraints are equally significant. High upfront capital costs, uncertain long-term profitability, and the absence of proven business models for eSAF production mean that investment decisions would be difficult to justify under current market conditions. This is also made complicated by a structural contracting mismatch, eSAF producers require long-term offtake agreements of 10–15 years to regain their capital investments, while airlines, paying close attention to the volatility of fossil fuel markets, typically prefer short-term supply contracts of 1–3 years. This mismatch creates a financing gap that prevents technically viable projects from reaching final investment decision.

Another constraint to pay particular attention to is the cross-border logistics challenge for both hydrogen and CO₂ which is critical to eSAF supply chain viability. The cheapest locations to produce green hydrogen in Europe, areas with high renewable energy density such as the Iberian Peninsula, close to North Africa, or Scandinavia, are rarely co-located with the industrial centres or airports where eSAF would be consumed. Transporting hydrogen across borders is technically complex and costly and introduces energy losses and additional cost. CO₂ logistics present a parallel challenge. Biogenic CO₂, the preferred near-term feedstock for eSAF synthesis given its lower cost compared to DAC, is concentrated at specific industrial point sources such as biogas upgrading plants, cement factories, and waste incinerators. These sources are geographically fixed and may not be located near either the renewable electricity supply or the PtL plant. Transporting captured CO₂ requires compression or liquefaction, specialised pipelines or tanker infrastructure. (Transport & Environment, 2025; CATF, 2026).

The infrastructural constraints extend further to the final delivery stage. Insufficient airport fuel logistics for large-scale SAF distribution, limited hydrogen transport and storage networks, and the absence of dedicated CO₂ distribution infrastructure across the EU

collectively mean that even where eSAF is produced, getting it reliably to the point of use remains a logistical challenge.

Putting all these together, these constraints make clear that eSAF deployment is a multi-sector integration challenge ranging from energy production, chemical processing, cross-border feedstock logistics, and aviation fuel distribution.

5.3 Capital Intensity and Investment Requirements

The production of eSAF through power-to-liquid pathways is highly capital-intensive due to the integration of multiple process units within a single system. These include renewable electricity supply, electrolyzers for hydrogen production, carbon capture systems, synthesis reactors (e.g., Fischer–Tropsch or Methanol-to-Jet), and downstream refining and storage infrastructure. Estimates from the European Commission suggest that achieving the scale of renewable and low-carbon fuel production required under EU transport decarbonisation policies could require billions of euros in investments (European Commission, 2024). Individual eSAF plants may require capital expenditures ranging from several hundred million euros to over €1 billion depending on plant size, configuration, and location. This high capital intensity creates a significant barrier to entry, particularly for first-of-a-kind projects, which are subject to higher financing costs due to perceived technological and market risks.

5.4 Infrastructure Requirements: Plants and Energy Systems

Meeting eSAF demand requires not only financial investment but also the physical deployment of industrial infrastructure at scale. Based on an estimated demand of 2.3 Mt/year by 2035, the number of required production facilities depends on plant capacity. Assuming plant sizes between 100 kt/year and 500 kt/year, Europe would require approximately 5 to 23 commercial-scale eSAF plants. This implies the development of multiple industrial clusters rather than isolated facilities.

In addition to synthesis plants, eSAF production is strongly dependent on renewable electricity. PtL processes typically require on the order of 18–25 MWh of electricity per tonne of fuel produced, depending on pathway efficiency (IEA, 2023). This level of electricity demand is significant and without sufficient low-cost renewable electricity, large-scale eSAF production remains economically unviable.

5.5 Transition from Feasibility to Risk

Meeting ReFuelEU targets may be technically conceivable, but it also reveals the scale of the challenge. The transition from theoretical feasibility, lab demonstrations or pilot demonstrations to large-scale deployment introduces multiple layers of uncertainty. Even with strong policy support, the deployment of eSAF at scale is exposed to a range of economic, technological, and regulatory risks that may delay or constrain development.

5.6 Key Economic and Systemic Risks

Financing Risk: One of the most critical barriers to eSAF deployment is the difficulty of securing financing for capital-intensive projects. Large-scale eSAF plants require substantial

upfront investment, typically financed through a combination of equity and debt. However, lenders require predictable revenue streams to reduce risk exposure. In the aviation sector, fuel procurement is traditionally based on short-term contracts linked to volatile fossil fuel prices. This creates a mismatch with the long-term offtake agreements required to secure financing for eSAF projects. Without stable, long-term purchase agreements from airlines or policy-backed revenue guarantees, many projects may struggle to reach final investment decision (IEA, 2023).

Power Price Risk: The cost of renewable electricity is an important part of eSAF production cost. If electricity prices remain high or volatile, the resulting fuel will remain significantly more expensive than conventional jet fuel. Although renewable energy costs have declined over time, future price trajectories remain uncertain due to factors such as grid constraints, competing demand from other sectors, and market design. High electricity prices could therefore undermine the economic viability of eSAF projects, particularly in regions without abundant low-cost renewable resources (IRENA, 2022).

Technology and Scale-Up Risk: While the individual components of eSAF production including electrolysis, carbon capture, and fuel synthesis are technically proven, their large industrial scale integration remains relatively new. First-of-a-kind plants often face commissioning delays, operational inefficiencies, and cost overruns. In addition, scaling from pilot projects to larger facilities introduces additional complexity, including process integration challenges, catalyst performance issues, and system reliability concerns. These risks may slow the pace of deployment and increase project costs (IEA, 2023).

Carbon Source Risk: The availability of suitable carbon dioxide sources represents another structural uncertainty. In the short term, eSAF production may rely on concentrated industrial CO₂ streams from sectors such as cement, steel, or bioenergy. However, as these sectors decarbonise, such point sources may decline over time. Direct air capture (DAC) offers a long-term alternative but is currently associated with high costs and energy requirements. The transition from industrial CO₂ to DAC then introduces additional uncertainty in both cost and scalability (IEA, 2023).

Policy and Regulatory Risk: Although ReFuelEU provides a clear demand signal, the broader policy environment remains complex and evolving. Factors such as sustainability criteria, certification requirements, carbon accounting rules, and subsidy mechanisms can significantly affect project economics. Changes in policy design or delays in implementing supporting instruments create uncertainty for investors. Given the long lifetimes of eSAF assets, regulatory stability is essential for project viability.

Overall, we see that the economic analysis highlights a major point in the ReFuelEU regulation. On one hand, regulatory mandates create strong and predictable demand. On the other hand, the supply side depends on a complex combination of capital investment, infrastructure deployment, energy availability, and risk management.

Bridging this gap requires coordinated action across multiple sectors, including energy, industry, finance, and transport. Without such coordination, the pace of supply development may fall short of policy targets.

CONCLUSION

The ReFuelEU Aviation Regulation represents a significant and necessary step toward decarbonising the aviation sector by creating a legally binding demand for sustainable fuels, including eSAF. From a strategic perspective, the regulation is well-founded, as it addresses one of the most difficult sectors to decarbonise and provides long-term visibility to investors.

However, the analysis presented in this study indicates that while the targets are technically achievable, their implementation is subject to substantial economic and systemic challenges. From a technical perspective, the targets are possible, particularly in the long term. From a near-term economic perspective, they are challenging. The 2030 and 2035 milestones may prove difficult if project development remains slow and production costs remain significantly above fossil jet fuel.

Also, the regulation is more ambitious on the demand side than on the supply side. While blending targets are legally defined, the industrial systems needed to meet them; renewable electricity generation, electrolyzers, CO₂ capture networks, synthesis plants, logistics infrastructure, and financing mechanisms, are not guaranteed at the same pace.

Therefore, the regulation can be considered realistic in direction, but optimistic in timing. The regulation can be viewed as pragmatic because aviation is one of the hardest sectors to decarbonise and requires a policy push. Yet it may need stronger complementary instruments to achieve its broader aim.

Overall, ReFuelEU Aviation should be viewed not as a complete solution, but as a necessary first framework whose success depends on multiple dimensions and factors. If implementation keeps pace with ambition, Europe could become a global leader in eSAF. If not, the result may be high compliance costs and increasing dependence on imported fuels.

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